

# IPL Batting Performance Analysis

## Business Requirement

### Business Scenario

The management team of the Indian Premier League (IPL) wants to analyze batting performances throughout the tournament to identify the most impactful batsmen, high-scoring teams, and batting trends across different matches and venues. Understanding these insights will help commentators, team analysts, coaches, and franchise owners evaluate player performance and improve future strategies.

As a **Junior Data Analyst**, your responsibility is to analyze the IPL ball-by-ball batting dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify top-performing batsmen, analyze team batting performance, study scoring patterns, and provide meaningful business recommendations.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and generate actionable insights.

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## Project Objective

Analyze the IPL batting dataset to:

- Identify the top-performing batsmen in the tournament.
  - Analyze team batting performance.
  - Study scoring patterns across overs and venues.
  - Understand batting contributions through boundaries and total runs.
  - Generate business insights and recommendations.
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## Dataset Description

The dataset contains **ball-by-ball batting information** from IPL matches, including match details, batting team, bowler, striker, runs scored, extras, wickets, and venue information.

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# Dataset Features

| Column           | Description              |
|------------------|--------------------------|
| Match ID         | Unique Match Identifier  |
| Season           | IPL Season               |
| Phase            | Tournament Stage         |
| Match No         | Match Number             |
| Date             | Match Date               |
| Venue            | Match Venue              |
| Batting Team     | Team Batting             |
| Bowling Team     | Team Bowling             |
| Innings          | Innings Number           |
| Over             | Over Number              |
| Striker          | Batsman Facing the Ball  |
| Bowler           | Bowler Name              |
| Runs of Bat      | Runs Scored from the Bat |
| Extras           | Extra Runs               |
| Wide             | Wide Balls               |
| Leg Byes         | Leg Bye Runs             |
| Byes             | Bye Runs                 |
| No Balls         | No Balls                 |
| Wicket Type      | Type of Dismissal        |
| Player Dismissed | Dismissed Player         |
| Fielder          | Fielder Involved         |

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## Tasks to Perform

## Task 1 – Data Loading

- Import the required libraries.
  - Load the dataset using Pandas.
  - Display the first five records.
  - Display the last five records.
  - Find the number of rows and columns.
  - Display all column names.
  - Check the data types of each column.
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## Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
  - Check duplicate records.
  - Display summary statistics.
  - Find the number of unique batsmen.
  - Find the number of unique bowlers.
  - Find the number of unique venues.
  - Find the number of batting teams.
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## Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
  - Handle missing values.
  - Convert the **Date** column into Date format.
  - Ensure **Runs of Bat**, **Over**, **Extras**, **Wide**, **No Balls**, **Leg Byes**, and **Byes** have the correct data types.
  - Replace missing values in wicket-related columns where necessary.
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## Business Questions

Answer the following questions using Python and Pandas.

### Batsman Analysis

- Which batsman scored the highest total runs in the tournament?
  - Display the Top 10 batsmen based on total runs scored.
  - Which batsmen faced the highest number of deliveries?
  - Which batsmen hit the most fours?
  - Which batsmen hit the most sixes?
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## Team Batting Analysis

- Which batting team scored the highest total runs?
  - Display the total runs scored by each team.
  - Which team scored the highest runs in a single match?
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## Match Analysis

- Which matches produced the highest total runs?
  - Which venues hosted the highest-scoring matches?
  - Which tournament phase recorded the highest total runs?
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## Over Analysis

- Which over generated the highest number of runs?
  - Compare runs scored across all overs.
  - Which overs were the most productive for batsmen?
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## Venue Analysis

- Which venue hosted the highest number of matches?
  - Which venue recorded the highest total runs?
  - Which venues were most favorable for batting?
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## Wicket Analysis

- How many wickets fell during the tournament?
- Which wicket type occurred most frequently?
- Which batsmen were dismissed the highest number of times?

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## Extras Analysis

- What is the total number of extra runs?
  - Which type of extra (Wide, No Ball, Bye, Leg Bye) occurred most frequently?
  - Which matches recorded the highest number of extras?
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## Data Visualization

Create **at least five charts** using Matplotlib.

### Mandatory Charts

- Top 10 Batsmen by Total Runs (Bar Chart)
- Total Runs by Batting Team (Bar Chart)
- Runs Scored Across Overs (Line Chart)
- Distribution of Runs per Delivery (Histogram)
- Wicket Type Distribution (Pie Chart)

### Optional Chart

- Top 10 Venues by Total Runs (Horizontal Bar Chart)
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## Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which batsmen consistently contributed the most runs?
  - Which teams had the strongest batting performance?
  - Which overs were the highest scoring?
  - Which venues favored batting the most?
  - Which type of dismissal occurred most frequently?
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## Business Recommendations

Provide **at least 3 recommendations** for the IPL team management.

Example recommendations:

- Build batting strategies around the tournament's most consistent run scorers.
  - Analyze high-scoring venues while planning match strategies.
  - Strengthen batting performance during overs that traditionally produce fewer runs.
  - Improve batting partnerships by studying successful batting combinations and scoring patterns.
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## Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
  - Cleaned Dataset (if modified)
  - Report
  - Recommendations (if any)
  - 5-minute PowerPoint Presentation
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## Evaluation Criteria

| Criteria                   | Marks      |
|----------------------------|------------|
| Data Loading & Exploration | 15         |
| Data Cleaning              | 15         |
| Data Analysis              | 30         |
| Visualizations             | 20         |
| Business Insights          | 10         |
| Recommendations            | 10         |
| <b>Total</b>               | <b>100</b> |